

ADM - RecSys 2025

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Challenge Overview

- Goal: Build **Universal Behavioral Profiles** (user representations).
- Input data: *user events* (e.g., purchases, cart events, page views, searches).
- Submissions: **user embeddings only**.
- Evaluation: Performed on multiple **downstream tasks**.

Key Points

- One set of embeddings $\Rightarrow$ used for all tasks.
- Models trained separately per task using submitted embeddings.
- Final score = sum of performances across all tasks.
- Both **open** and **hidden tasks** used in evaluation.

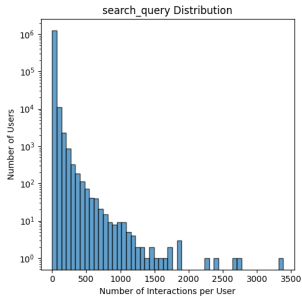
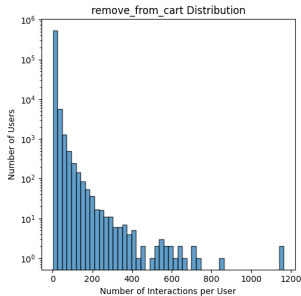
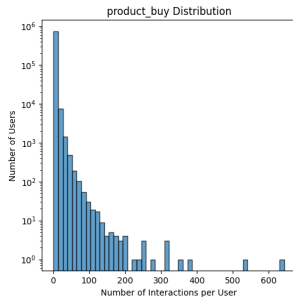
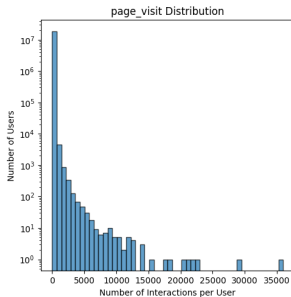
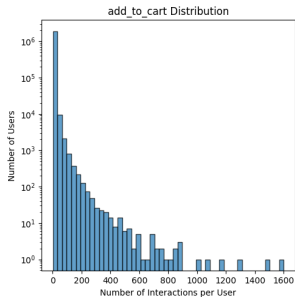
- **Churn Prediction:** Binary classification into 1: user will churn or 0: user will not churn. Churn task is performed on a subset of active users with at least one `product_buy` event in history.
Task name: `churn`
- **Categories Propensity:** Multi-label classification into one of 100 possible labels. The labels represent the 100 most often purchase product categories.
Task name: `propensity_category`
- **Product Propensity:** Multi-label classification into one of 100 possible labels. The labels represent the 100 most often purchase products in train target data.
Task name: `propensity_sku`

$$\text{Score} = 0.8 \times \text{AUROC} + 0.1 \times \text{Novelty} + 0.1 \times \text{Diversity}$$

- The metric is computed on the validation set consisting the last 14 days of the dataset.

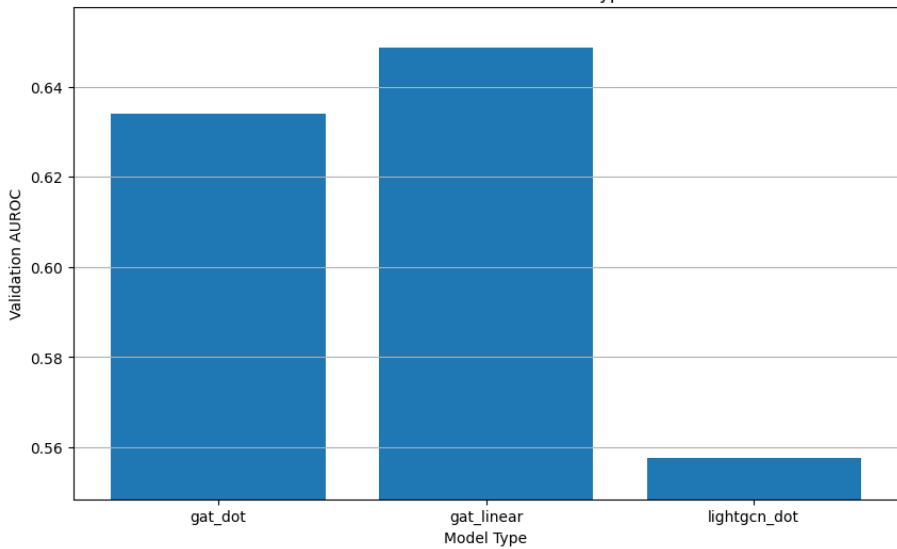
- **Data types:** Product purchases, cart additions/removals
- **Preprocessing:** Filtered users with ≥ 5 interactions
- **Temporal split:**
 - Input events: Historical data
 - Train targets: 2 weeks preceding validation
 - Validation targets: Final 2 weeks
- **Simplification:** User-category interactions (vs. individual items)
- **Scale:** 21,233 users, 578,701 events, 6,384 item categories

FULL DATASET: Distribution of Interactions per User

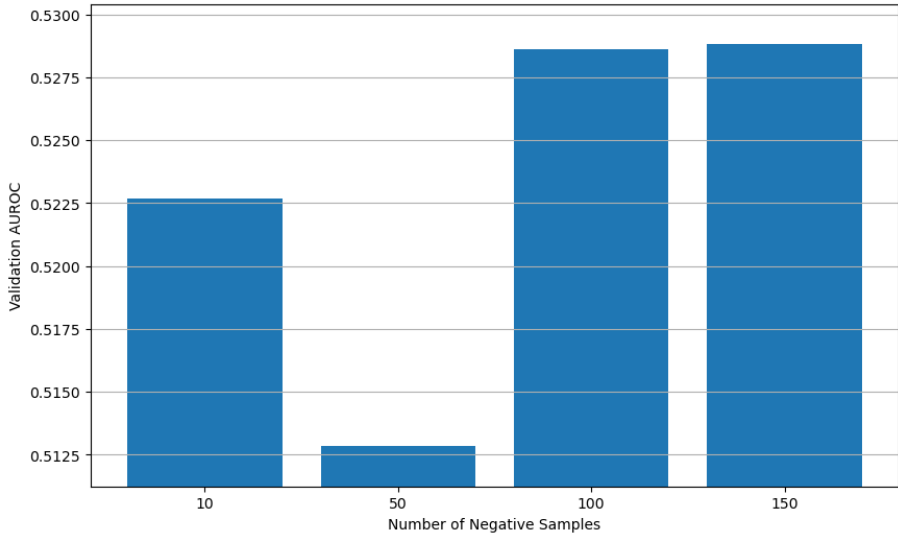


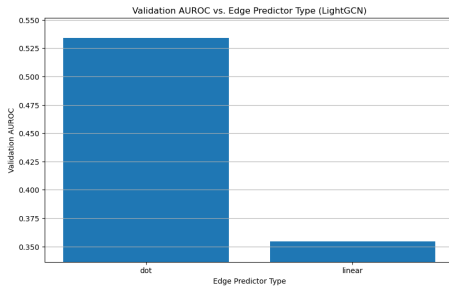
- **Model:** Graph model applied to the bipartite user interaction graph. Compared two variants:
 - LightGCN
 - Graph Attention Transformer
- **BPR Loss:** Bayesian Personalized Ranking - optimizes pairwise ranking between positive and negative user-item interactions
- **User Embeddings:** Extracted from final GNN layer outputs for downstream task evaluation
- **Interaction Scoring:** Tried two variants:
 - Dot product similarity
 - Linear layer transformation

Validation AUROC vs. Model Type

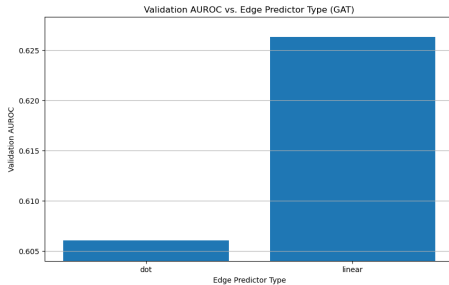


Validation AUROC vs. Number of Negative Samples (LightGCN)

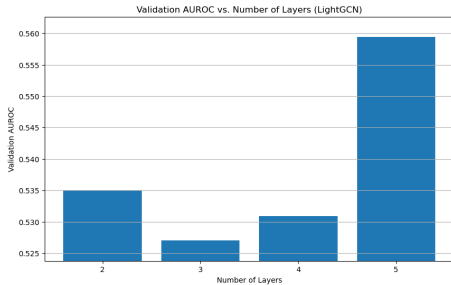




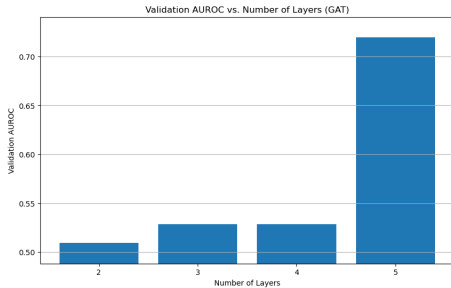
LightGCN – Edge Type



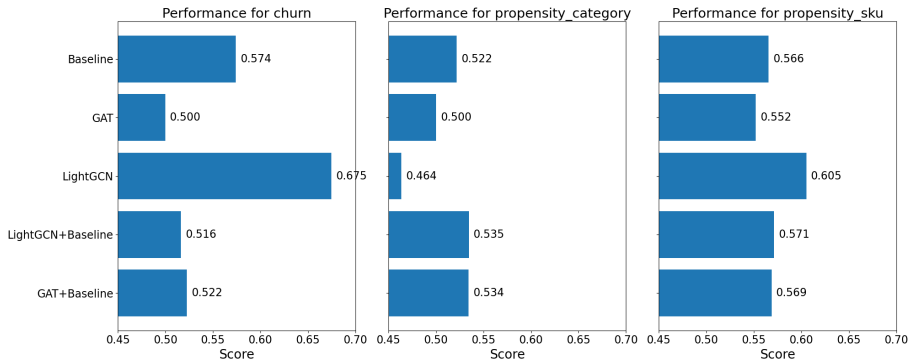
GAT – Edge Type



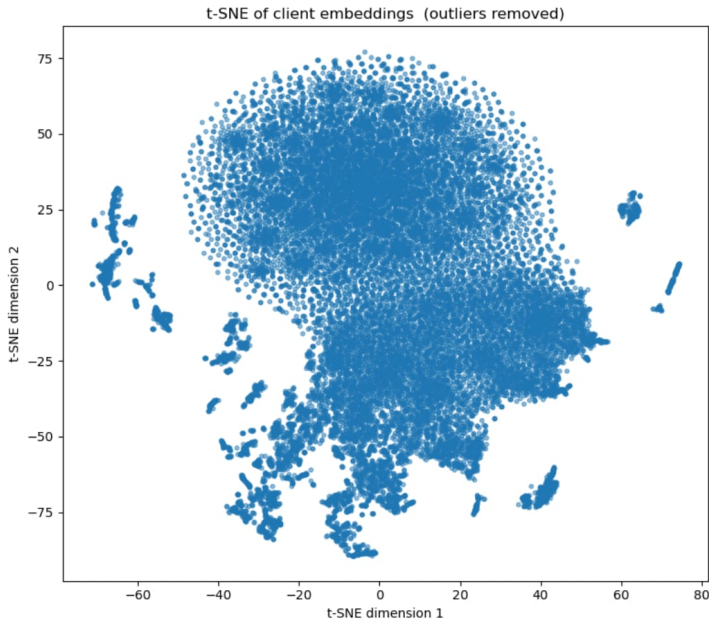
LightGCN – Number of Layers



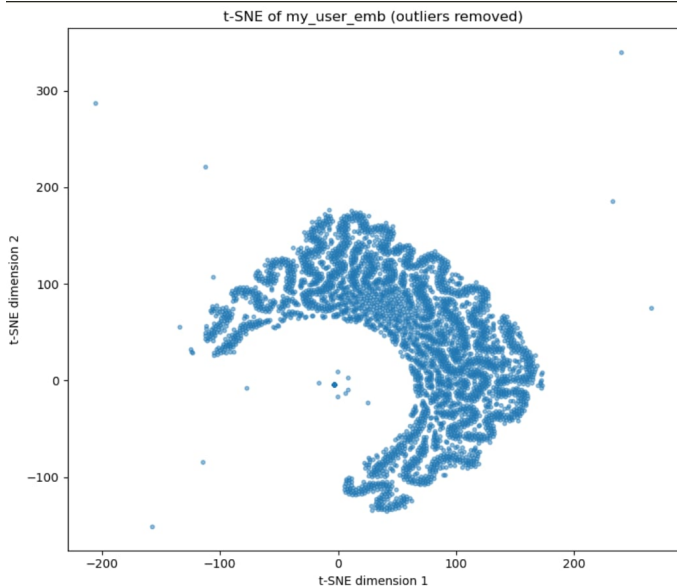
GAT – Number of Layers



LightGCN embeddings



GAT embeddings



Baseline embeddings

